03

Lecture 03

2G/3G Core Networks

شبکه‌های هستهٔ نسل دوم و سوم

2G — **3G** — 4G — 5G — 6G

COURSE PART
Part I – 2G/3G foundations (Lectures 1–4)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 1–2; spread spectrum basics; decibels

In this lecture

1 Learning objectives . . . 2
2 Prerequisite knowledge . . . 2
3 Introduction and motivation . . . 3
4 Fundamental concepts . . . 3
5 Detailed technical explanation . . . 4
6 Mathematical models . . . 6
7 Network architecture and diagrams . . . 8
8 Step-by-step procedures . . . 8
9 Worked examples . . . 9
10 Technology comparison . . . 11
11 Key takeaways . . . 11
12 Exercises . . . 12
13 Further reading . . . 13

Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-03-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Explain** the requirements that drove 3G (IMT-2000) and why a new radio interface, WCDMA, was needed.
- **LO2** **Draw** the UMTS architecture – UE, NodeB, RNC, CS and PS core domains – and name the interfaces Uu, Iub, Iur, Iu-CS and Iu-PS.
- **LO3** **Compute** processing gain, the uplink load factor and the noise rise of a WCDMA cell, and **estimate** its voice pole capacity.
- **LO4** **Describe** the HSPA evolution (Releases 5–8) and **derive** the HSDPA peak rate from codes, spreading factor and modulation.
- **LO5** **Trace** an RRC connection and radio access bearer (RAB) set-up with RANAP.
- **LO6** **Explain** the role of IMS and **evaluate** why LTE removed the CS/PS split.

2 Prerequisite knowledge

This lecture builds directly on the GSM/GPRS architecture of Lecture 1 and the circuit-switched core of Lecture 2. You should also recall that a spread-spectrum signal occupies a bandwidth much larger than its data rate, and that a receiver correlating with the right code recovers the signal while other codes look like noise.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
UMTS	سامانهٔ جهانی مخابرات موبایل	The 3GPP 3G system: WCDMA radio + evolved GSM core.
WCDMA	دسترسی چندگانهٔ تقسیم کد پهن‌باند	3G radio interface: 5 MHz carrier, 3.84 Mchip/s.
NodeB	نود بی	3G base station: radio transmission, spreading, inner-loop power control.
RNC	کنترل‌گر شبکهٔ رادیویی	Controls NodeBs: radio resources, handover, ciphering.
Spreading factor (SF)	ضریب گسترش	Chips per symbol; trades rate against protection.
Processing gain	بهرهٔ پردازش	W/R: how much despreading lifts the wanted signal.
Soft handover	تحویل نرم	UE connected to several cells at once (make-before-break).
Noise rise	افزایش نویز	Increase of uplink interference over thermal noise.
RAB	حامل دسترسی رادیویی	Bearer between UE and core network, set up by RANAP.
HSPA	دسترسی بستهٔ پرسرعت	Shared-channel evolution of WCDMA (HSDPA + HSUPA).
IMS	زیرسامانهٔ چندرسانه‌ای IP	SIP-based control of IP multimedia services.

3 Introduction and motivation

By the late 1990s GSM had proved that mobile telephony could be global, but it could not become a data network. EDGE squeezed about 59 kbit/s out of a timeslot, and one 200 kHz carrier can never give a single user megabits per second. The ITU's IMT-2000 programme asked for up to 2 Mbit/s indoors, 384 kbit/s for pedestrians and 144 kbit/s for vehicles, flexible multiplexing of services with different rates on one connection, and global roaming.

3GPP's answer, UMTS, combined two decisions. On the radio side it adopted a completely new interface, Wideband CDMA, with a 5 MHz carrier. On the network side it did the opposite: it *reused* the GSM/GPRS core – MSC, HLR, SGSN, GGSN – and connected it to the new radio network through a new interface, Iu. This lecture studies that hybrid: a new radio access network attached to an evolved 2G core with two parallel domains, and the strains that eventually led LTE to abandon the circuit-switched domain altogether.

4 Fundamental concepts

4.1 Code division multiple access

In WCDMA all users of a cell transmit at the same time on the same 5 MHz carrier. Each user's symbols are multiplied by a code running at the chip rate $W = 3.84\,\text{Mchip/s}$. The receiver multiplies again by the same code (despreading): the wanted signal is concentrated back into its narrow data bandwidth, while signals with other codes remain spread and act like noise.

Definition 1 • Spreading factor and processing gain

The spreading factor is the number of chips per symbol, $\text{SF} = W/R_\text{s}$. For a service with information rate R, the processing gain is $G_\text{p} = W/R$. It is the factor by which despreading increases the signal-to-interference ratio.

Two families of codes are used. *Channelisation codes* (orthogonal variable spreading factor, OVSF) separate the channels sent by one transmitter; *scrambling codes* separate transmitters (cells in the downlink, UEs in the uplink). OVSF codes form a tree (Figure 4); a code can be used only if no ancestor or descendant is in use, so assigning a short (low-SF, high-rate) code blocks a whole branch.

4.2 Interference-limited capacity

Because everyone shares one carrier, WCDMA capacity is limited by interference, not by a fixed number of channels: every additional user raises the interference seen by all the others. Two consequences follow. First, *power control* is essential: an uplink user close to the NodeB must not drown distant users (the near–far problem), so the NodeB adjusts each UE's power 1500 times per second. Second, capacity is *soft*: the operator can trade a little quality for more users, and the cell coverage shrinks as the load grows (cell breathing).

4.3 Soft handover

Since neighbouring cells use the same frequency, a UE near a cell border can be connected to two or three cells simultaneously. In the uplink the RNC selects the best frame; in the downlink the UE combines the signals of all cells in its active set. Soft handover avoids the gap of a hard handover and provides macro-diversity, at the cost of extra NodeB and Iub resources. It is analysed in Lecture 4.

5 Detailed technical explanation

5.1 UTRAN: NodeB and RNC

The UMTS Terrestrial Radio Access Network (UTRAN) [6] has two node types. The **NodeB** performs the physical layer: channel coding, spreading, modulation, RF and the fast inner-loop power control. The **RNC** (Radio Network Controller) is much more powerful than a GSM BSC. It terminates the radio protocols RLC, MAC (for dedicated channels) and RRC, performs admission control, handover decisions, outer-loop power control, macro-diversity combining, and – unlike GSM, where ciphering ended at the BTS – ciphering and integrity protection. An RNC plays different roles for a given UE:

- **SRNC** (serving RNC): terminates the Iu connection and the RRC signalling of that UE;
- **DRNC** (drift RNC): lends radio resources of its cells when the UE is in soft handover with cells of another RNC; data passes over *Iur* to the SRNC;
- **CRNC** (controlling RNC): the RNC that controls a given NodeB's resources.

5.2 Interfaces

Table 1. UMTS interfaces.

Interface	Between	Control protocol	Notes
Uu	UE – NodeB/RNC	RRC	WCDMA radio interface
Iub	NodeB – RNC	NBAP	Transport originally ATM, later IP
Iur	RNC – RNC	RNSAP	Soft handover across RNCs; no GSM equivalent
Iu-CS	RNC – MSC	RANAP	Replaces the A interface; AMR speech
Iu-PS	RNC – SGSN	RANAP	User plane in GTP-U; replaces Gb

A key design change is that the Iu-PS user plane uses GTP-U all the way from the RNC, so the SGSN no longer handles LLC/SNDCP towards the terminal; the radio-specific protocols end in the RNC. RANAP [5] carries the signalling for RAB management, relocation of the SRNC, paging and security mode control.

5.3 The core network: CS and PS domains

The UMTS core keeps the two domains of GSM/GPRS:

- The **circuit-switched (CS) domain** (MSC/VLR, GMSC) serves voice and video calls with Erlang-dimensioned circuits. From Release 4 the MSC is split into an *MSC Server* (call control and mobility, the control plane) and a *Media Gateway* (MGW, the user plane that switches and transcodes the speech). This allows the CS core transport to run over IP or ATM – an early example of control/user plane separation.
- The **packet-switched (PS) domain** (SGSN, GGSN) serves IP traffic with PDP contexts, now with QoS classes (conversational, streaming, interactive, background).

The HLR/AuC is shared; the AuC now produces *quintets* (RAND, XRES, CK, IK, AUTN) for mutual authentication (*UMTS AKA*, [1]). AUTN lets the USIM verify that the challenge came from its home network and is fresh; IK protects signalling integrity. Lecture 11 derives these values.

5.4 From Release 99 to HSPA

Release 99 carried packet data on dedicated channels with fixed spreading factors and 10–80 ms transmission time intervals (TTIs). Allocating a dedicated code to a bursty user wasted code space, and scheduling in the RNC was slow. HSPA moved intelligence to the NodeB:

- **Release 5 – HSDPA** [2]: a shared downlink channel (HS-DSCH) using up to 15 codes of SF 16, a 2 ms TTI, adaptive modulation (QPSK/16QAM), fast hybrid ARQ and channel-dependent scheduling in the NodeB, driven by CQI reports from the UE every 2 ms.
- **Release 6 – HSUPA** (enhanced uplink): an enhanced dedicated channel (E-DCH) with NodeB scheduling and HARQ, up to 5.76 Mbit/s.

- **Release 7 – HSPA+**: 64QAM in the downlink, 2×2 MIMO, 16QAM uplink, and a flat-architecture option.
- **Release 8**: dual-carrier HSDPA (two adjacent 5 MHz carriers).

The philosophical shift matters more than the numbers: the network stopped allocating fixed resources to users and started scheduling a shared pipe every 2 ms according to channel quality – exactly the paradigm LTE adopts from the start (Lecture 7).

5.5 IMS: IP multimedia over the PS domain

The IP Multimedia Subsystem (IMS), introduced in Release 5 [3], defines how to offer voice, video and messaging as IP services controlled by SIP [8]. Its core elements are the call session control functions: the **P-CSCF** (proxy, the UE's first contact point), the **I-CSCF** (interrogating, entry point of the home network) and the **S-CSCF** (serving, which registers the user and executes service logic), plus the HSS, the evolution of the HLR. In 3G, IMS was rarely used for mass-market voice; its real role came with LTE, where it is the only way to make calls (VoLTE, Lecture 5).

6 Mathematical models

6.1 Processing gain and the required SINR

Problem. What signal-to-interference ratio must a WCDMA receiver see before despreading for a given service quality?

Model. Required energy per bit to noise-plus-interference density $(E_\mathrm{b}/N_0)_\mathrm{req}$, chip rate W, service bit rate R.

Derivation. Despreading accumulates W/R chips per bit, so

$$\frac{E_\mathrm{b}}{N_0} = \frac{W}{R} \cdot \mathrm{SINR}_\mathrm{wideband} \quad \Longrightarrow \quad \mathrm{SINR}_\mathrm{req} = \frac{(E_\mathrm{b}/N_0)_\mathrm{req}}{G_\mathrm{p}}, \qquad G_\mathrm{p} = \frac{W}{R}. \tag{1}$$

Example and interpretation. AMR voice at $R = 12.2\,\mathrm{kbit/s}$: $G_\mathrm{p} = 3.84 \times 10^6 / 12.2 \times 10^3 = 315$ (25.0 dB). With $(E_\mathrm{b}/N_0)_\mathrm{req} = 5\,\mathrm{dB}$, the wideband SINR may be as low as $5 - 25 = -20\,\mathrm{dB}$: the signal is buried 20 dB below the interference and still decoded. A 384 kbit/s data user has only $G_\mathrm{p} = 10\,\mathrm{dB}$ and needs roughly 15 dB more received power – which is why high-rate users consume a large share of the cell's capacity.

6.2 Uplink load factor, noise rise and pole capacity

Problem. How many simultaneous users can a WCDMA cell carry in the uplink?

Assumptions. N users of the same service, perfect power control, received powers adjusted so that each user just meets its E_b/N_0; voice activity factor ν; other-cell interference equal to a fraction i of own-cell interference; thermal noise power P_N [7].

Derivation. Let I_{tot} be the total received wideband power and P_j the received power of user j. Its requirement is

$$\left(\frac{E_\text{b}}{N_0}\right)_j = \frac{W}{\nu_j R_j}\,\frac{P_j}{I_{\text{tot}} - P_j} \quad\Longrightarrow\quad P_j = L_j\, I_{\text{tot}}, \qquad L_j = \frac{1}{1 + \dfrac{W}{(E_\text{b}/N_0)_j\, R_j\, \nu_j}}. \tag{2}$$

Summing own-cell users and adding other-cell interference, $I_{\text{tot}} = (1+i)\sum_j P_j + P_\text{N}$, which defines the uplink load factor

$$\eta_\text{UL} = (1+i)\sum_{j=1}^{N} L_j \qquad \text{and} \qquad \frac{I_{\text{tot}}}{P_\text{N}} = \frac{1}{1-\eta_\text{UL}}. \tag{3}$$

The ratio $I_{\text{tot}}/P_\text{N}$ is the *noise rise*. It diverges as $\eta_\text{UL} \to 1$; the number of users at which this happens is the *pole capacity*

$$N_\text{pole} = \frac{1}{(1+i)L} = \frac{1 + \dfrac{W}{(E_\text{b}/N_0)R\nu}}{1+i}. \tag{4}$$

Interpretation. Figure 1 shows the noise rise: 3 dB at 50% load, 6 dB at 75%, 10 dB at 90%. Each decibel of noise rise is a decibel of lost link budget – coverage shrinks as the cell fills up. Operators therefore plan for 50–75% load, not for the pole.

Limitations. The model assumes perfect power control, a single service and a fixed i; real systems need margins for power-control errors and fast fading.

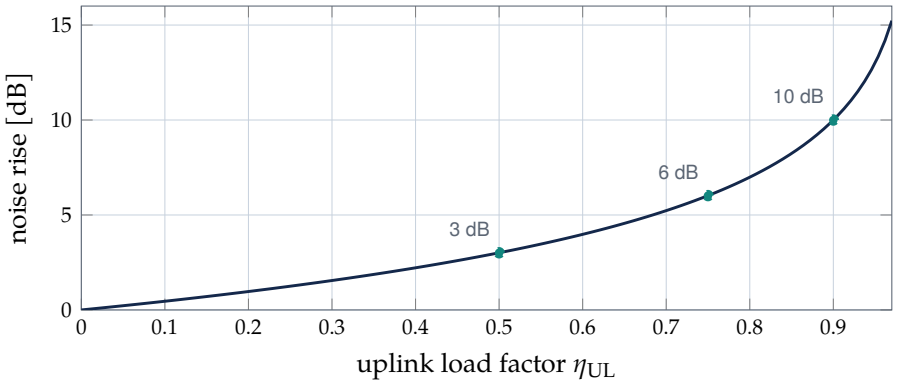


Figure 1. Uplink noise rise $-10\log_{10}(1-\eta_\text{UL})$ from Eq. (3).

6.3 HSDPA peak rate

With n_c codes of spreading factor SF and b bits per symbol, the symbol-level rate of the HS-DSCH is

$$R_\text{peak} = n_\text{c}\,\frac{W}{\text{SF}}\,b\,r_\text{c}, \tag{5}$$

where r_c is the effective code rate. For $n_\text{c} = 15$, SF $= 16$, 16QAM ($b = 4$) and $r_\text{c} \to 1$: $15 \times 240\,\text{ksymbol/s} \times 4 = 14.4\,\text{Mbit/s}$; the highest standardised UE category reaches 14.0 Mbit/s. 64QAM gives 21.6 Mbit/s (21.1 in practice), and two MIMO layers or two carriers double it (Figure 2). These are peak values for one user alone in a cell with excellent SINR; average cell throughput is several times lower.

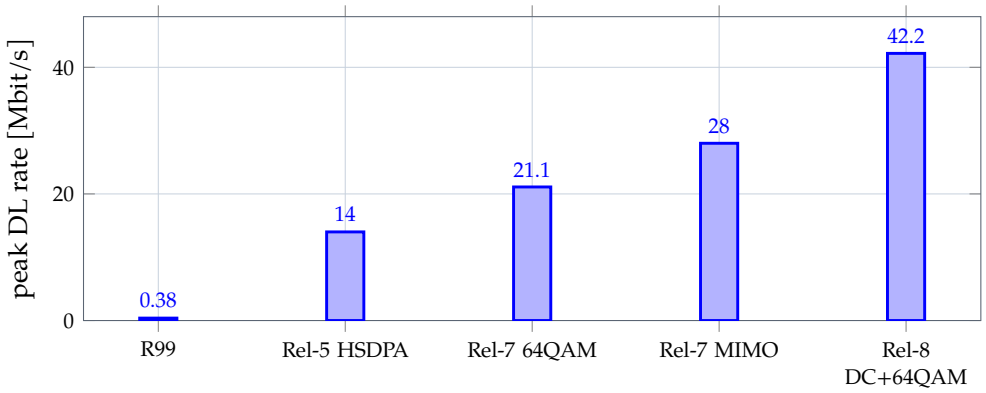


Figure 2. Peak downlink rates along the WCDMA/HSPA evolution (R99 shown with its practical 384 kbit/s; categories with the highest standardised rate).

7 Network architecture and diagrams

Figure 3 shows the UMTS network. Compared with GSM (Lecture 1, Figure 3) three things changed: the RAN uses new nodes and interfaces (NodeB, RNC, Iub, Iur); the core attaches to the RAN only through Iu, which has a CS and a PS flavour; and the PS user plane is a GTP-U tunnel that starts at the RNC. The core network itself is recognisably the GSM/GPRS core.

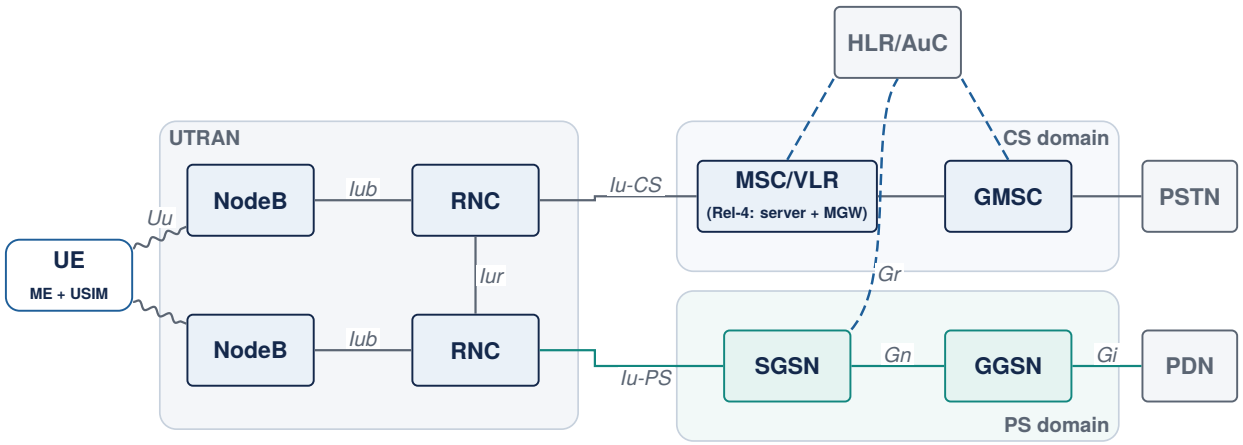


Figure 3. UMTS architecture with the UTRAN, the circuit-switched and the packet-switched core domains. Dashed: MAP signalling to the HLR.

8 Step-by-step procedures

Procedure 1 • PS service request and RAB establishment

1. The idle UE sends RRC Connection Request on the random access channel.
2. The RNC performs admission control, sets up a radio link in the NodeB (NBAP) and answers RRC Connection Setup; the UE confirms with RRC Connection Setup Complete.
3. The UE sends a NAS Service Request to the SGSN, carried in an RRC Initial Direct Transfer

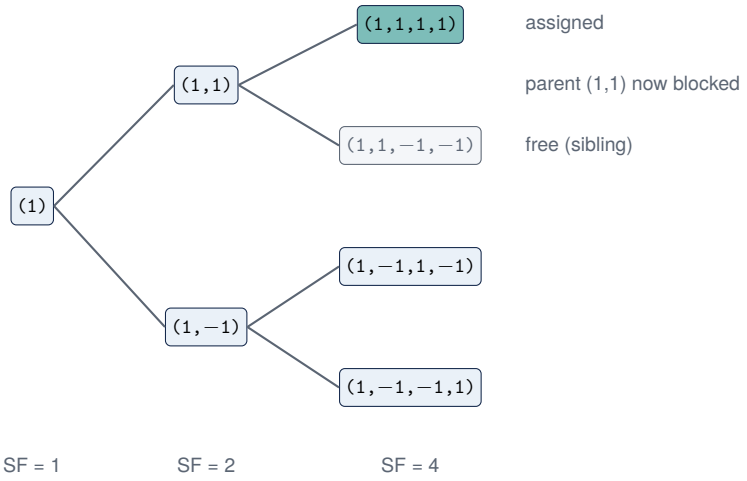


Figure 4. OVSF code tree. Assigning $(1,1,1,1)$ blocks its parent $(1,1)$ and the root, but its sibling $(1,1,-1,-1)$ remains usable; all codes of one SF are mutually orthogonal.

and the RANAP Initial UE Message.

4. The SGSN may run UMTS AKA; it then sends RANAP Security Mode Command with CK and IK to the RNC, which starts ciphering and integrity protection towards the UE.
5. The SGSN requests the bearer with RANAP RAB Assignment Request, including QoS and its GTP-U tunnel endpoint.
6. The RNC configures the radio bearer with RRC Radio Bearer Setup; the UE answers Radio Bearer Setup Complete; the RNC returns RAB Assignment Response with its own tunnel endpoint. User data now flows over the radio bearer and the Iu-PS tunnel.

Note the layering: NAS messages (Service Request) travel transparently from UE to SGSN; access-stratum signalling (RRC) ends in the RNC; RANAP connects the two worlds. This split between *non-access stratum* and *access stratum* remains in LTE and 5G.

9 Worked examples

Worked example 1 • Voice pole capacity

Problem. AMR voice, $R = 12.2\,\text{kbit/s}$, $(E_b/N_0)_{\text{req}} = 5\,\text{dB}$, activity $\nu = 0.67$, other-cell interference ratio $i = 0.65$. Find the pole capacity and the number of users at 50% load.

Solution. $W/((E_b/N_0)R\nu) = 3.84\times10^6/(3.162\times12\,200\times0.67) = 148.6$. Equation (4): $N_{\text{pole}} = (1+148.6)/1.65 = 90.6$. Each user loads the cell by $(1+i)L = 1.65/149.6 = 1.10\%$, so 50% load corresponds to about 45 users, with a noise rise of 3 dB.

Interpretation. 45 voice users per 5 MHz carrier and sector, i.e. 9 per MHz – a substantial gain over GSM's roughly 94 channels in 10 MHz with reuse (Lecture 1).

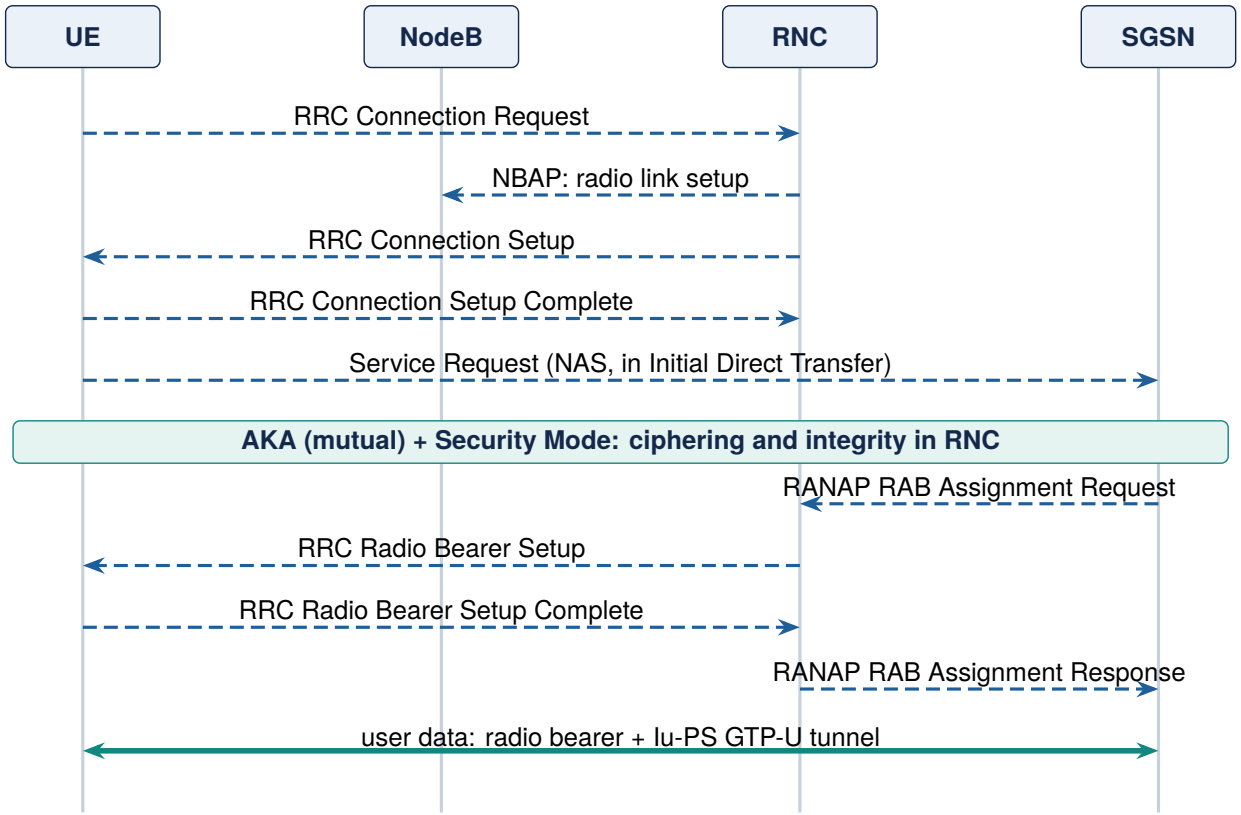


Figure 5. RRC connection and RAB set-up for a packet service (Procedure 1).

Worked example 2 • Mixing services

Problem. In the cell of Example 1, one 384 kbit/s data user needs $(E_b/N_0) = 1.5\,\text{dB}$ with $\nu = 1$. How many voice users does it replace?

Solution. $W/((E_b/N_0)R\nu) = 3.84 \times 10^6/(1.413 \times 384\,000) = 7.08$, so $L = 1/8.08 = 0.124$ and its load contribution is $1.65 \times 0.124 = 20.4\%$. One voice user contributes 1.10%: the data user replaces about 18 voice users.

Interpretation. Resource consumption in CDMA scales with $R \cdot E_b/N_0$, which is why high-rate packet data on dedicated channels was so costly in Release 99.

Worked example 3 • OVSF code budget

Problem. HSDPA reserves 15 codes of SF 16. What fraction of the downlink code tree is left for everything else?

Solution. SF 16 has 16 codes; 15 are used, leaving one SF 16 branch, i.e. $1/16 = 6.25\%$ of the tree, which must hold common channels (pilot, broadcast, paging), the HS-SCCH control channels and any Release 99 dedicated channels.

Interpretation. In practice operators configure 5–10 HS-PDSCH codes and keep the rest for voice – code space, not power, often limits the downlink.

10 Technology comparison

Table 2. Mobile generations up to 3.5G (indicative peak values).

	GSM/GPRS	EDGE	UMTS R99	HSPA (Rel-5/6)
Multiple access	TDMA/FDMA	TDMA/FDMA	WCDMA	WCDMA + shared channel
Carrier	200 kHz	200 kHz	5 MHz	5 MHz
Peak DL	171 kbit/s	474 kbit/s	384 kbit/s (2 Mbit/s theor.)	14 Mbit/s
Handover	Hard	Hard	Soft + hard	Soft (DCH); serving-cell change (HS-DSCH)
Scheduling	BSC/PCU	BSC/PCU	RNC, 10–80 ms	NodeB, 2 ms
Core	CS + PS	CS + PS	CS + PS (Iu)	CS + PS (Iu)
Authentication	One-way (triplet)	One-way	Mutual AKA (quintet)	Mutual AKA

The CS/PS split and why LTE removed it. Running two core domains means two sets of nodes, two mobility management procedures (location area and routing area updates), two charging systems and two sets of interconnection agreements. Every innovation (for example a new QoS scheme) had to be done twice. Once HSPA proved that a shared packet channel can carry voice-quality traffic, the economic case for a single, packet-only core became overwhelming: LTE's EPC (Lecture 9) has no CS domain, and voice is carried by IMS over IP. The price is complexity elsewhere – QoS bearers must now guarantee what a circuit guaranteed for free, and legacy voice requires fallback procedures (CS fallback, SRVCC) during the transition.

11 Key takeaways

Key takeaways

- UMTS = new radio (WCDMA, 5 MHz, 3.84 Mchip/s) + evolved GSM/GPRS core, joined by Iu.
- The RNC terminates RLC/MAC/RRC, ciphering and integrity; SRNC/DRNC roles enable soft handover across RNCs over Iur.
- CDMA capacity is interference-limited: $\eta_{\mathrm{UL}} = (1+i)\sum L_j$, noise rise $1/(1-\eta_{\mathrm{UL}})$, plan for 50–75% load.
- Processing gain W/R lets low-rate users operate at negative SINR; high-rate users cost proportionally more.
- HSPA moved scheduling and HARQ to the NodeB with a 2 ms TTI – the template for LTE.
- Mutual authentication (AKA) and integrity protection fixed the main GSM security gaps.

- Two core domains doubled cost and complexity; LTE kept only the packet domain and moved voice to IMS.

12 Exercises

Exercise 1 • Level 1 – conceptual • RNC versus BSC

List three functions performed by the UMTS RNC that the GSM BSC did not perform, and explain which design choice of WCDMA made each necessary.

Exercise 2 • Level 1 – conceptual • Why Iur?

GSM has no interface between BSCs. Explain why UMTS needs Iur, and what would happen to soft handover at an RNC border without it.

Exercise 3 • Level 2 – numerical • Load budget

Using the parameters of Example 1, how many voice users can be admitted if the noise rise must not exceed 4 dB? How does the answer change if i drops to 0.4 (better isolated cells)?

Exercise 4 • Level 2 – numerical • HSDPA rate

A cell configures 10 HS-PDSCH codes. Compute the symbol-level peak rate for QPSK, 16QAM and 64QAM, and the rate with code rate $r_c = 0.75$ for 16QAM.

Exercise 5 • Level 3 – analytical • Cell breathing

Assume a path-loss exponent of 3.5. By what factor does the cell radius shrink when the load grows from 50% to 75%, if the UE transmit power is the limiting factor? Derive the general expression for the radius ratio as a function of the two load factors.

Exercise 6 • Level 3 – analytical • Mixed traffic

A cell must carry N_v voice users (parameters of Example 1) and N_d users at 64 kbit/s with $E_b/N_0 = 2$ dB and $\nu = 1$, at a total load of 60%. Derive the linear constraint between N_v and N_d and plot (or tabulate) the admissible region.

Solutions

Solution to Exercise 1

(1) Macro-diversity combining and soft-handover control – needed because all cells use the same frequency. (2) Outer-loop power control (setting the SIR target from frame error rates) – needed because CDMA is interference-limited and power control is the main resource control. (3) Ciphering and integrity protection, terminating RLC/MAC/RRC – moved from BTS/SGSN to keep security and retransmission in one place despite soft handover across many NodeBs. Admission control based on load (Eq. (3)) is a fourth, because capacity is soft.

Solution to Exercise 2

In soft handover a UE may be connected to cells of two RNCs. Without Iur the SRNC could not receive the uplink frames decoded by the other RNC's NodeB, nor send downlink data through it; the UE would have to perform a hard handover at every RNC border, losing macro-diversity exactly where links are weakest. Iur lets the DRNC forward frames to the SRNC, which keeps the Iu connection unchanged until a relocation is worthwhile.

Solution to Exercise 3

Noise rise 4 dB: $1-\eta = 10^{-0.4} = 0.398$, so $\eta_{\max} = 0.602$. With $(1+i)L = 1.10\%$, $N = 0.602/0.0110 = 54$ users. With $i = 0.4$: $(1+i)L = 1.4/149.6 = 0.936\%$, $N = 64$ users – better cell isolation (sectorisation, antenna tilt) translates directly into capacity.

Solution to Exercise 4

Per code: $W/16 = 240$ ksymbol/s; 10 codes: 2.4 Msymbol/s. QPSK: 4.8 Mbit/s; 16QAM: 9.6 Mbit/s; 64QAM: 14.4 Mbit/s. 16QAM with $r_c = 0.75$: $9.6 \times 0.75 = 7.2$ Mbit/s.

Solution to Exercise 5

The extra noise rise between loads η_1 and η_2 is $\Delta = 10\log_{10}\frac{1-\eta_1}{1-\eta_2}$ dB. With a fixed UE power the tolerable path loss drops by Δ; for $L_p \propto d^n$, $d_2/d_1 = 10^{-\Delta/(10n)} = \left(\frac{1-\eta_2}{1-\eta_1}\right)^{1/n}$. For $\eta_1 = 0.5$, $\eta_2 = 0.75$: $\Delta = 3.01$ dB and $d_2/d_1 = 0.5^{1/3.5} = 0.820$: the radius shrinks by 18% and the area by 33%.

Solution to Exercise 6

Data user: $W/((E_b/N_0)R) = 3.84\times10^6/(1.585\times64\,000) = 37.9$, $L_d = 1/38.9 = 0.0257$. Constraint: $1.65\,(N_v \times 0.00668 + N_d \times 0.0257) \le 0.60$, i.e. $0.0110\,N_v + 0.0424\,N_d \le 0.60$. Corner points: $(N_v, N_d) = (54, 0)$, $(0, 14)$; e.g. 30 voice users leave room for 6 data users. One 64 kbit/s user costs about 3.9 voice users.

13 Further reading

Holma and Toskala [7] is the standard reference for WCDMA and HSPA radio planning, including the load equations of Section 6.2. The UTRAN architecture is in [6], RANAP in [5], HSDPA in [2] and 3G security in [1]. IMS is specified in [3] on top of SIP [8]; the general 3GPP architecture is in [4].

References

[1] 3GPP, *3G security; security architecture*, TS 33.102, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/33102.htm.

[2] 3GPP, *High speed downlink packet access (HSDPA); overall description; stage 2*, TS 25.308, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/25308.htm.

[3] 3GPP, *IP multimedia subsystem (IMS); stage 2*, TS 23.228, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23228.htm.

[4] 3GPP, *Network architecture*, TS 23.002, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/23002.htm.

[5] 3GPP, *UTRAN Iu interface radio access network application part (RANAP) signalling*, TS 25.413, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/25413.htm.

[6] 3GPP, *UTRAN overall description*, TS 25.401, 3rd Generation Partnership Project. [Online]. Available: https://www.3gpp.org/DynaReport/25401.htm.

[7] H. Holma and A. Toskala, *WCDMA for UMTS: HSPA Evolution and LTE*, 5th ed. Wiley, 2010.

[8] J. Rosenberg, H. Schulzrinne, G. Camarillo, *et al.*, "SIP: Session initiation protocol", IETF, RFC 3261, 2002. [Online]. Available: https://www.rfc-editor.org/rfc/rfc3261.